

1 Causal 4D Memory

What was observed, when, and with what evidence

last positive view



entity ID + state versions + provenance

RECALL

2 Predictive Belief

Is the last state still valid? If not, where next?

$$\rho = P(\text{stay} \mid H_o, t_q)$$

Last seen

$b = 0.18$

Relocate

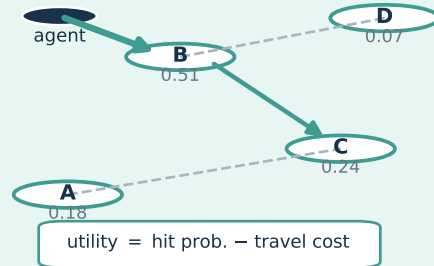
$b = 0.82$



PREDICT

3 Belief-Guided Route

Choose the best success--cost trade-off



ACT

4 Observe, Update, Replan

Turn positive and negative views into new belief

Inspect B

target not seen; coverage = 0.91

$$b^+(s) \propto b^-(s)P(z \mid s)$$

visibility-aware negative evidence

Posterior: C becomes first

write evidence → replan

VERIFY